

Lecture 2: Functions, Graphs, and Transformations

A function assigns exactly one output to each allowed input. Precalculus studies a handful of basic function families and the operations — transformation, composition, and inversion — that connect them. Once the parent graphs are known, most other graphs are just shifted, scaled, or reflected versions of them.

1. Functions, Parent Graphs, and Symmetry

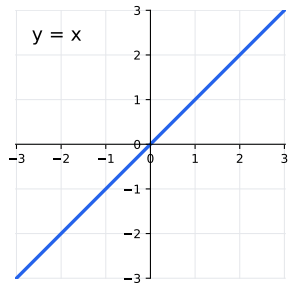
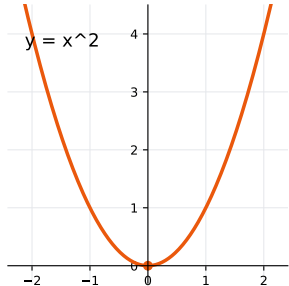
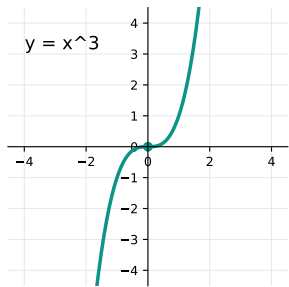
Definition. A function f assigns to each input x in its **domain** exactly one output $f(x)$; the **range** is the set of outputs produced. The graph of f is the set of points $(x, f(x))$.

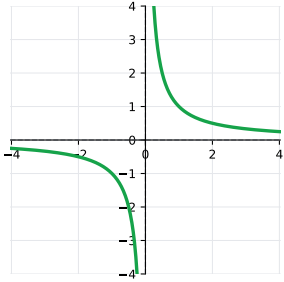
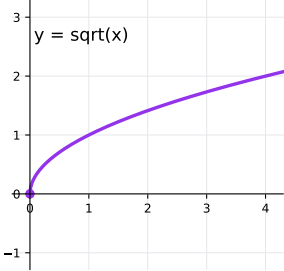
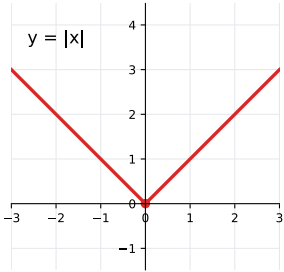
Vertical line test. A graph represents a function of x if and only if every vertical line meets it at most once.

Example (the unit circle is not a function). The circle $x^2 + y^2 = 1$ fails the vertical line test: for $-1 < x < 1$ it contains both $y = \sqrt{1 - x^2}$ and $y = -\sqrt{1 - x^2}$. Solving for y splits the circle into two functions of x , the upper and lower semicircles.

Example (a dropped object). If an object is dropped from height H at time $t = 0$, its height in meters is $h(t) = H - 4.9t^2$, ignoring air resistance. Until it reaches the ground, $0 \leq t \leq \sqrt{H/4.9}$, each time determines exactly one height, so h is a function of time. Its graph is the descending part of a parabola.

Six basic functions are worth knowing by their domain, range, and graph.

Function	Domain	Range	Graph
$y = x$	$(-\infty, \infty)$	$(-\infty, \infty)$	
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$	
$y = x^3$	$(-\infty, \infty)$	$(-\infty, \infty)$	

Function	Domain	Range	Graph
$y = 1/x$	$x \neq 0$	$y \neq 0$	
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$	
$y = x $	$(-\infty, \infty)$	$[0, \infty)$	

Definition. f is **even** if $f(-x) = f(x)$ for every x in its domain (graph symmetric about the y -axis); f is **odd** if $f(-x) = -f(x)$ (graph symmetric about the origin).

Example. Of the six parent functions, $y = x^2$ and $y = |x|$ are even; $y = x$, $y = x^3$, and $y = 1/x$ are odd; $y = \sqrt{x}$ is neither, since its domain is not symmetric about 0.

Exercise 1. Find the domain and range of $f(x) = \frac{\sqrt{x-1}}{x-3}$.

Exercise 2. True or false, with a reason:

- the range of $y = x^2$ is all real numbers
- the graph of $y = 1/x$ crosses the y -axis
- $y = \sqrt{x}$ is the top half of $y = x^2$ turned on its side
- $y = x^3$ has domain restricted to $x \geq 0$
- $|x|$ and x^2 have the same range

Exercise 3. Decide whether each is a function of x , and whether it is even, odd, or neither:

- $y = 3x - 2$
- $x = y^2 + 1$
- $y = x^4$
- $y = x^3 + x^2$

Exercise 4 (harder). Find the domain and range of $g(x) = \frac{1}{\sqrt{x^2 - 4}}$.

2. Transformations and Polynomial Families

Suppose (a, b) lies on $y = f(x)$.

New function	Point on new graph	Effect
$f(x) + k$	$(a, b + k)$	shift up k
$f(x - h)$	$(a + h, b)$	shift right h
$-f(x)$	$(a, -b)$	reflect across x -axis
$f(-x)$	$(-a, b)$	reflect across y -axis
$af(x)$	(a, ab)	vertical scale by a

Exercise 5. Describe the transformations from the parent function in each, then sketch the resulting graph:

- $y = -3|x + 2| + 1$
- $y = \sqrt{x - 4} - 2$
- $y = \frac{2}{x + 1} + 3$

Exercise 6. If $(2, -1)$ is on the graph of f , find the corresponding point on $y = f(x) + 4$, on $y = f(x - 3)$, and on $y = -f(-x)$.

Linear functions. Applying the table to $y = x$ gives $y = ax + b$: a vertical scale by a (the slope) followed by a shift of b (the intercept).

Exercise 7. Find the line through $(-1, 4)$ and $(3, -8)$.

Quadratic functions. Every quadratic is a transformation of the parent graph $y = x^2$, built up one step at a time from the table above:

1. Start with $y = x^2$, vertex $(0, 0)$.
2. Scale vertically by a : $y = ax^2$. The vertex stays at $(0, 0)$; the parabola narrows if $|a| > 1$, widens if $|a| < 1$, and flips upside down if $a < 0$.
3. Shift right h : $y = a(x - h)^2$. The vertex moves to $(h, 0)$.
4. Shift up k : $y = a(x - h)^2 + k$. The vertex moves to (h, k) .

None of these steps change the parabola's shape, so $y = a(x - h)^2 + k$ is always a parabola with vertex (h, k) , opening upward if $a > 0$ and downward if $a < 0$.

A quadratic given in standard form $ax^2 + bx + c$ is the same parabola, just not yet written to show h and k . **Completing the square** finds them: factor a out of the first two terms, then add and subtract the constant that completes the square inside the parentheses.

$$ax^2 + bx + c = a\left(x^2 + \frac{b}{a}x\right) + c = a\left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}\right) - \frac{b^2}{4a} + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}.$$

Comparing with $a(x - h)^2 + k$ gives

$$h = -\frac{b}{2a}, \quad k = c - \frac{b^2}{4a}.$$

Example. For $2x^2 - 8x + 3$, $a = 2$ and $b = -8$, so $h = -\frac{-8}{2(2)} = 2$ and $k = 3 - \frac{(-8)^2}{4(2)} = 3 - 8 = -5$. Thus $2x^2 - 8x + 3 = 2(x - 2)^2 - 5$, vertex $(2, -5)$ — matching the four-step construction: start from x^2 , scale by 2 (narrower, still opening up), shift right 2, shift down 5.

Exercise 8. Sketch $y = -2(x + 1)^2 + 3$, then state its vertex, axis, opening direction, and range.

Exercise 9. Rewrite $y = 3x^2 - 12x + 7$ in vertex form.

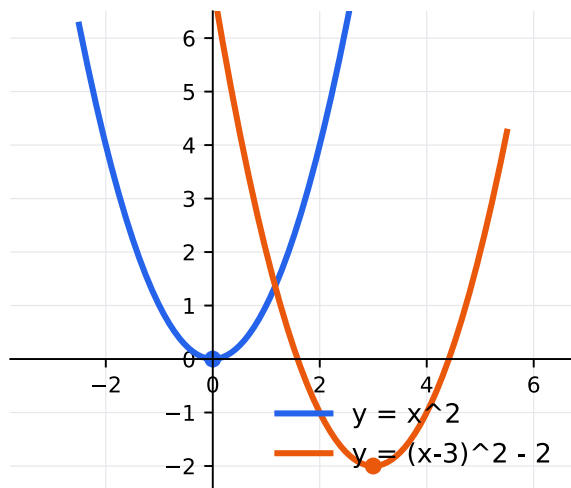


Figure 1: Parabola and its shift

Cubic functions. Applying the table to $y = x^3$ gives $y = a(x - h)^3 + k$, a shift of the inflection point to (h, k) . Since x^3 is strictly increasing, $a(x - h)^3 + k$ is strictly increasing when $a > 0$ and strictly decreasing when $a < 0$; either way it has **no turning points**.

Not every cubic has this form. The cubic $y = x^3 - x = x(x - 1)(x + 1)$ has three real zeros, -1 , 0 , and 1 ; between consecutive zeros the graph must turn around, so it has two turning points. No shift or scale of $y = x^3$ can produce this shape.

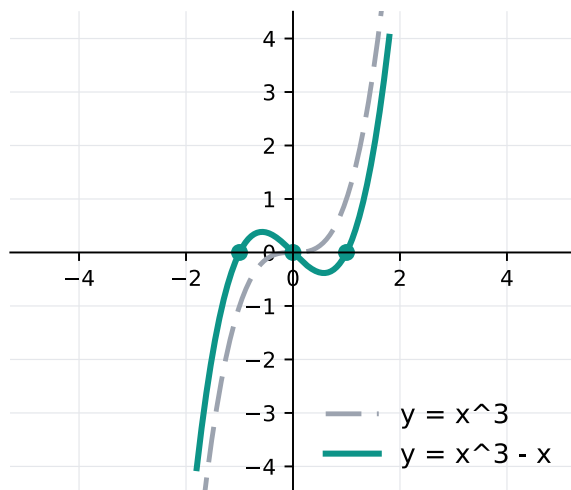


Figure 2: Cubic parent graph compared with x cubed minus x , which has turning points

Exercise 10. Is $y = (x + 1)^3 - 4$ a transformation of $y = x^3$? Is $y = x^3 - 3x$? Explain.

3. Composite Functions

Definition. The **composite function** $f \circ g$ applies g first and then f :

$$(f \circ g)(x) = f(g(x)).$$

Its domain consists of the inputs in the domain of g for which $g(x)$ is in the domain of f . In general, $f \circ g \neq g \circ f$.

Example. Let $f(x) = \sqrt{x}$ and $g(x) = 3 - x$. Then $(f \circ g)(x) = \sqrt{3 - x}$, with domain $(-\infty, 3]$, whereas $(g \circ f)(x) = 3 - \sqrt{x}$, with domain $[0, \infty)$. The order changes both the formula and the allowed inputs.

Exercise 11. Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Exercise 12. Let $f(x) = \sqrt{x}$ and $g(x) = x^2 - 4$. Find the formula and domain of $(f \circ g)(x)$ and $(g \circ f)(x)$.

4. Increasing, Decreasing, and Extrema

Definition. f is **increasing** on an interval if $x_1 < x_2$ in that interval always gives $f(x_1) < f(x_2)$; f is **decreasing** if it always gives $f(x_1) > f(x_2)$. A function that is increasing (or decreasing) on its entire domain is automatically one-to-one, since it never repeats an output — the key fact behind the next section.

A point where f changes from increasing to decreasing is a **local maximum**; a change from decreasing to increasing is a **local minimum**. A **global** (or absolute) maximum or minimum is the largest or smallest value of f over its whole domain.

Example. Reading off the parent graphs:

- $y = x$ is increasing everywhere, with no extrema.
- $y = x^2$ is decreasing on $(-\infty, 0]$ and increasing on $[0, \infty)$, with a global minimum 0 at $x = 0$.
- $y = x^3$ is increasing everywhere, with no extrema.
- $y = 1/x$ is decreasing on $(-\infty, 0)$ and decreasing on $(0, \infty)$ — but, as Exercise 15 shows, it is *not* decreasing on its full domain.
- $y = \sqrt{x}$ is increasing on $[0, \infty)$, with a global minimum 0 at $x = 0$.
- $y = |x|$ is decreasing on $(-\infty, 0]$ and increasing on $[0, \infty)$, with a global minimum 0 at $x = 0$.

For $y = a(x - h)^2 + k$, the vertex is a global minimum if $a > 0$ and a global maximum if $a < 0$; the function decreases before $x = h$ and increases after it (or the reverse, if $a < 0$).

Exercise 13. For $y = 3(x - 2)^2 - 5$, state the intervals of increase and decrease and the minimum value.

Exercise 14 (harder). The cubic $y = x^3 - 3x = x(x - \sqrt{3})(x + \sqrt{3})$ is an odd function with zeros at $-\sqrt{3}$, 0, and $\sqrt{3}$. Using only the end behavior and the zeros (no calculus): decide whether the graph is increasing or decreasing immediately to the left of $x = -\sqrt{3}$ and immediately to the right of $x = \sqrt{3}$; then explain why there must be a local maximum strictly between $x = -\sqrt{3}$ and $x = 0$, and (using the odd symmetry of the graph) a local minimum strictly between $x = 0$ and $x = \sqrt{3}$.

Exercise 15 (Challenge). Show that $f(x) = 1/x$ is decreasing on $(0, \infty)$ and decreasing on $(-\infty, 0)$, but give specific numbers $x_1 < x_2$ (with x_1 negative and x_2 positive) showing that f is *not* decreasing on its full domain.

5. Inverse Functions

Definition. f has an **inverse function** f^{-1} exactly when, for every y in the range of f , the equation $y = f(x)$ can be solved for x uniquely — each output comes from exactly one input. (Graphically, this is the **horizontal line test**; algebraically, f is called **one-to-one**.) The inverse satisfies

$$f^{-1}(f(x)) = x, \quad f(f^{-1}(x)) = x.$$

The domain of f^{-1} is the range of f , and vice versa. To find f^{-1} : write $y = f(x)$, swap x and y , and solve for y — if the solution is not unique, f has no inverse on that domain.

Example. For $f(x) = 3x - 7$: swapping gives $x = 3y - 7$, so $f^{-1}(x) = \frac{x + 7}{3}$.

Example (restricted domain). For $f(x) = x^2$, the equation $y = x^2$ does not solve uniquely for x on all of $\mathbb{R}$: both $x = \sqrt{y}$ and $x = -\sqrt{y}$ work when $y > 0$. Restricting to $x \geq 0$ makes the solution unique, $x = \sqrt{y}$, so $f^{-1}(x) = \sqrt{x}$ there.

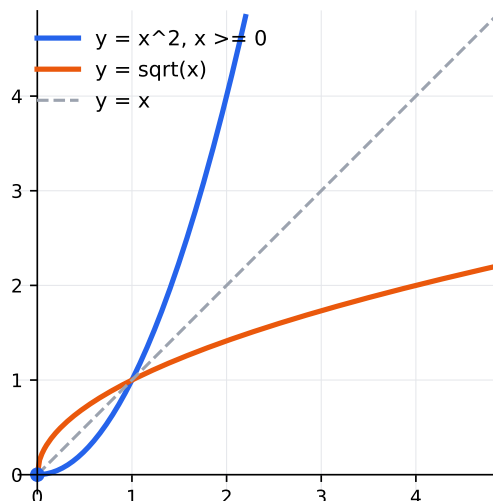


Figure 3: y equals x squared for x at least 0 and its inverse, the square root function

Theorem. The graph of f^{-1} is the reflection of the graph of f across $y = x$.

Proof. If (a, b) is on the graph of f , then $b = f(a)$, so $a = f^{-1}(b)$ and (b, a) is on the graph of f^{-1} . The point (b, a) is the reflection of (a, b) across $y = x$. ■

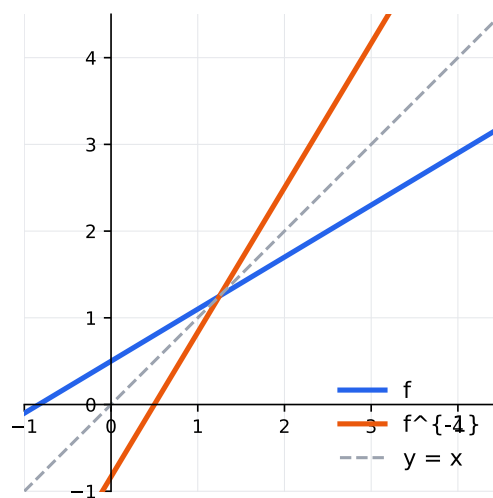


Figure 4: A function and its inverse, reflected across $y=x$

Exercise 16. For each, decide whether $y = f(x)$ can be solved uniquely for x on the stated domain (that is, whether f is one-to-one there):

- $f(x) = 2x + 1$ on $\mathbb{R}$
- $g(x) = x^3$ on $\mathbb{R}$
- $g(x) = \frac{1}{x^2}$ on $\mathbb{R}$

Exercise 17. Find the inverse of $f(x) = \frac{x-1}{x+2}$ and state its domain and range.

Exercise 18 (Challenge). Let f be one-to-one and $g(x) = af(b(x-h)) + k$ with $a, b \neq 0$. Find g^{-1} in terms of f^{-1} .

Exercise 19 (Challenge). Prove directly from the definitions that $(f^{-1})^{-1} = f$.

Exercise 20 (Challenge). Prove that if f is strictly increasing on an interval, then f is one-to-one there; and if f^{-1} exists, prove that f^{-1} is also strictly increasing.